Genesis: Institutional Trading & Strategy Intelligence Platform

Factual product brief and system specification for the Genesis platform. Component, schema, and dependency details live in the companion documents in this folder. Operational procedures live in backend/README.md and the root README.md.

Source of truth: Application code (`backend/app/`, `frontend/src/`, `scripts/`, `trades/`), Alembic revisions, SQLite WAL schemas, and `.env.example`.

Analysis & System Specification Date: 2026-10-30.

---

1. Problem Statement & Evolution

A retail options strategy idea typically arrives as a video, a social media post, a web article, or a Markdown note. Transforming that unstructured concept into a mathematically backtested, forward-tested strategy with realistic Indian options charges, reviewing it trade-by-trade, and deploying it into live market conditions has historically been fragmented, manual, and prone to execution slippage.

Genesis resolves this by providing an institutional quantitative pipeline spanning **Strategy Research & Lab**, **Automated Review & Ingestion**, **Real-Time Market Structure Analytics**, and **5 Autonomous Trading Crews** running continuous live operations.

---

2. Core Platform Capabilities

1. **Strategy Lab & Multi-Source Ingestion**:

- Ingests trading strategies from YouTube, Instagram, web documents (HTML), NotebookLM, or Markdown.

- Extracts structured Markdown with an executable `rules:` block.

- Detects specification gaps and asks clarifying questions, iteratively re-extracting with author responses.

2. **Deterministic Review Crew**:

- Executes a 7-step LangGraph StateGraph: syntax audit, parameter scenario building, historical backtesting with Indian options charges (brokerage, STT, exchange fees, GST, SEBI, stamp duty, slippage), hard-stop capital sweeps (2% to 5%), trade-by-trade post-mortems, verdict synthesis, and improvement proposals.

3. **5 Specialized Autonomous Trading Crews**:

- **PreMarketCrew (08:30 – 09:14 IST)**: Gap analysis, Call/Put gamma walls, Max Pain, PCR, event risk screening, and morning battleplan generation.

- **ScannerCrew (09:15 – 15:30 IST)**: Real-time multi-timeframe delta polling (1m, 2m, 5m, 15m), VWAP proxy, dynamic level projection, ATR deadband/lunch-lull filtering, and multi-strategy pattern detection.

- **TradeCrew (Live Market Hours)**: Macro trend slope evaluation, dynamic R:R (+1.8R to +4.5R), whale volume absorption trap veto, 3-Tier sizing, strictly OTM strike selection, Telegram ticket dispatch, and trade journaling.

- **ExecutionCrew**: Smart routing tranches (30% market / 70% limit chase), broker bridge (Paper and Live), and zero-drift position reconciliation.

- **PostMarketCrew (15:31+ IST)**: Intraday tape reconstruction, exit audits, walk-forward parameter sweeps, 50% Capital Safety Wall profit locking, code quality audit, and daily executive digest.

4. **Non-Repainting Market Structure Engine**:

- Mathematical swing high/low detection (2L/2R), Market Structure Break (`BOS`), Change of Character (`CHoCH`), Fair Value Gaps (`FVG`), dynamic trendlines, and session VWAP.

5. **Institutional Risk & Sizing Engine**:

- **3-Tier Execution Framework**: 3 Lots 1-OTM (standard momentum), 4 Lots 2-OTM (super-trend runner), and 2 Lots ATM (scalp mode).

- **50% Capital Safety Wall**: Permanently locks 50% of cumulative net realized profits into a protected floor, eliminating the risk of giving back gains.

- **Daily Circuit Breakers**: Max 3 daily trades, soft pause on 2 consecutive losses, mandatory hard stop on 3 losses.

6. **Single Source of Truth (SSOT) & Excel Sync**:

- High-throughput SQLite WAL database (`data/trades.db`) provides atomic, concurrent ACID transactions.

- Automatically synchronizes institutional Excel workbooks under `trades/` (`live_and_forward_trades.xlsx`, `todays_live_trades.xlsx`, and `by_strategy/`).

7. **Cryptographic Provenance & Dual Audit Chains**:

- Strategies approved for live deployment freeze `content_sha256` and `rules_sha256`, verified on every tick.

- Dual SHA-256 hash chains (`audit_events` and `live_audit_events`) guarantee mathematical proof of tamper resistance via `GET /api/audit/verify`.

---

3. Documents in This Directory

| Document | Role & Description |

| :--- | :--- |

| CURRENT_ARCHITECTURE_AND_AGENT_FLOWS.md | Exhaustive specification of the 5 Autonomous Crews, operational lifecycles, and Mermaid flowcharts. |

| CURRENT_SYSTEM_ARCHITECTURE_ANALYSIS.md | Component-by-component architectural audit, endpoint mapping, and middleware stacks. |

| CURRENT_DATABASE_ARCHITECTURE.md | Schema definitions, table columns, indices, and constraints across all databases. |

| DATABASE_TABLE_REFERENCE.md | Detailed table reference, field descriptions, and read/write access patterns. |

| INSTITUTIONAL_SCALING_LAW_AND_SAFETY_WALL.md | Mathematical proof and implementation of the 50% Safety Wall and 3-Tier sizing model. |

| LIBRARIES_AND_DEPENDENCIES.md | Exhaustive dependency analysis of libraries imported across backend and frontend. |

| FINAL_REPORT.md | Comprehensive engineering and academic report conforming to NIIT guidelines. |

---

4. Visual Architectural Diagrams

All architectural diagrams are maintained as vector SVGs and rendered into high-resolution PNGs in diagrams/:

| Diagram | Description | Links |

| :--- | :--- | :--- |

| **Full Platform Architecture** | High-level system architecture showing Next.js UI, FastAPI, Daemons, Multi-Tier Storage, and Externals. | SVG · PNG |

| **5 Autonomous Crews Ecosystem** | The 5 specialized crews (PreMarket, Scanner, Trade, Execution, PostMarket) and their 21 agents. | SVG · PNG |

| **Live Scanner & Execution Flow** | Multi-timeframe polling, dynamic levels, pattern detection, sizing, safety wall, and post-market loop. | SVG · PNG |

| **Review Crew & Trade Crew Workflow** | Workflow comparison between Research Lab Review Crew and live deployment Trade Crew. | SVG · PNG |

| **Multi-Tier Database Architecture** | Schema relationships between Research DB, Live DB, TradeStore SSOT, and Market DB. | SVG · PNG |

| **Lab Ingest Flow** | Ingestion, gap clarification, backtest replay, and report generation sequence. | SVG · PNG |

| **Live Trading Gate Flow** | Policy validation, snapshot hashing, live execution gates, and position reconciliation. | SVG · PNG |

| **Paper Trading Flow** | Interactive chart execution, delayed quotes, Indian charge deduction, and paper position tracking. | SVG · PNG |

To regenerate PNGs, HTML, DOCX, and PDF exports from the Markdown sources, run:

```
python3 DocumentsAndDiagrams/export_docs.py
```

---

5. Getting Started & Operations

- Complete setup, port bindings, CLI commands, and environment variables are documented in the root README.md.
- Quick service management:

```bash

./scripts/genesis.sh start # Starts Backend (8841), Frontend (3000), and Live Scanner Daemon

./scripts/genesis.sh status # Inspect health and running process PIDs

./scripts/genesis.sh logs # Tail unified system logs

```